

Inference theory, MASA03/ÄMAD04 Formula sheet

Elementary properties for Gaussian data

If $X_1, \dots, X_n$ are i.i.d. $N(0, 1)$, then

$$\begin{aligned}\bar{X} &\in N(0, 1/n), \\ \sum_{i=1}^n X_i^2 &\in \chi^2(n), \\ \sum_{i=1}^n (X_i - \bar{X})^2 &\in \chi^2(n-1), \\ \bar{X} \text{ and } \sum_{i=1}^n (X_i - \bar{X})^2 &\text{ are independent.}\end{aligned}$$

Multiple linear regression

If y_i are independent observations from $N(m_i, \sigma^2)$, where

$$m_i = \theta_0 + \theta_1(x_{1i} - \bar{x}_{1\cdot}) + \dots + \theta_p(x_{pi} - \bar{x}_{p\cdot}), \quad i = 1, 2, \dots, n,$$

with $\bar{x}_{j\cdot} = \sum_{i=1}^n x_{ji}/n$, then

$$\begin{aligned}\hat{\theta} &= (X^T X)^{-1} X^T Y, \\ \hat{\theta} &\in N(\theta, V(\hat{\theta})), \\ V(\hat{\theta}) &= \sigma^2 (X^T X)^{-1}, \\ Q(\hat{\theta}) &= (Y - X\hat{\theta})^T (Y - X\hat{\theta}) \\ \frac{Q(\hat{\theta})}{\sigma^2} &\in \chi^2(n-p-1) \\ \hat{\theta} \text{ and } Q(\hat{\theta}) &\text{ are independent.}\end{aligned}$$

Relations between distributions

If $U \in N(0, 1)$ and $V \in \chi^2(f)$, and U and V are independent, then

$$\frac{U}{\sqrt{V/f}} \in t(f).$$

If $V_1 \in \chi^2(f_1)$ and $V_2 \in \chi^2(f_2)$, and V_1 and V_2 are independent, then

$$\begin{aligned}V_1 + V_2 &\in \chi^2(f_1 + f_2), \\ \frac{V_1/f_1}{V_2/f_2} &\in F(f_1, f_2).\end{aligned}$$